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# Design and Implementation of an Embedded Control System for the Interconnection of a DC Microgrid to the AC Main Network

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CONTENTS



**Abstract**—This paper presents the design of an embedded control system and its implementation on a DC-AC bidirectional converter for the interconnection of a DC Microgrid (MG) with the AC Main Network (MN). This control system was designed in LabVIEW and implemented on the NI myRIO platform, and it was used for the synchronization of the MG with the MN. Its main goal is to keep the stability of the DC bus when energy is supplied to or extracted from the MN. The MG is composed of two 500 W photovoltaic panel arrays (PAs), each with a Boost DC-DC converter to generate a DC bus of 190 V, and a DC-AC bidirectional converter that interconnects the MG with the MN. The control system was tested considering nonl-inear loads such as computers, commercial fluorescent lamps, and LED, with the aim to analyze the response of the system to real life conditions. Results show that the embedded control system is capable to stabilize the DC bus at a voltage level of 190 V ± 5% when interacting with the different non-linear loads connected to the MG. This is an improvement on the control of the voltage variation previously presented in the literature of 190 V ± 10%.

Keywords: DC Microgrid, embedded control, photovoltaic panel, bidirectional converter, DC bus, SOGI-FLL algorithm, LabVIEW, NI myRIO, non-linear loads, AC main network
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## 1. INTRODUCTION

With the increasing demand of electricity and the scarcity of fossil fuels, the use of renewable energy technologies is an important alternative for the energy industry. Additionally, the increase of pollution and greenhouse effects has pointed toward the use of these type of technologies [1]. Among all the renewable energy technologies available, those based on solar energy are the most promising ones because of the availability of sunlight and because they can be working in an autonomous manner, or in a non-autonomous manner connected to the Main Network (MN) [2].

If the solar energy technology (Panel Array, or PA) is working in a non-autonomous manner, the energy produced is transferred directly to the MN, which requires that the electricity of the MN and that of the PA are synchronized in phase and frequency; otherwise, the possibility that the PA is damaged may exist. Thus, it is necessary a real-time phase and frequency synchronization system to allow the transfer of electricity to the MN [3]. To overcome this problem, a Second Order Generalized Integrator (SOGI) together with a Frequency Locked Loop (FLL) are commonly used to synchronize the voltage of the MN with that of the PA. The SOGI-FLL performs well even during the presence of disturbances in the MN such as voltage distortions (sag and swell), harmonics, unbalance, and frequency variations. However, the SOGI-FLL shows problems during offset corrections. To overcome this problem, a modified SOGI-FLL algorithm is proposed in Patil and Patel [4]. A small signal modeling of the SOGI-FLL algorithm is presented in Yi *et al*. [5] to analyze the effects on the impedance generated by voltage perturbations. Another SOGI-FLL algorithm is proposed in Singh *et al*. [6], the goal of this algorithm is to control the transfer of the energy generated by a 3 kW PA to the MN when a linear or non-linear load demand energy from the system.

The SOGI-FLL algorithm is also been implemented on embedded systems using different platforms, such as the Simulink-Matlab [7], FPGA of Xilinx Spartan 3E. An experimental fuzzy controller is used in Kumar *et al*. [8] to tune in the overall gain of an adaptive control approach based on a novel multilayer fifth-order generalized integrator algorithm. The experimental design of an embedded control system in a Spartan 3E card is proposed in Reddy *et al*. [9] using a power of 229 W, an open loop, AC linear and non-linear loads, and a sinusoidal signal equivalent to that of the MN that is generated in the FPGA. A 1 kW experimental prototype is presented in Yang *et al*. [10] where the SOGI-FLL control system and the algorithm for the Maximum Power Point Tracking (MPPT) are implemented in the DSP2812.

For three-phase systems, a control method based on the SOGI-FLL algorithm is proposed in Chittora *et al*. [11] to filter the harmonics, the phase angle extraction, and the extraction of the positive sequence voltage, for a DC distribution system that uses a three-phase AC-DC converter, the algorithm was implemented in the dSPACE 1104. Also, experimental results of a real-time performance of an embedded system applied to a three-phase rectifier that connects the MN with a 200 V DC bus are presented in Dagbagi *et al*. [12], showing the stability of the bus when connecting and disconnecting a 10 Ω and 2.5 A resistive load, the embedded system was implemented in a low-cost Xilinx Zynq FPGA SoC device. Also, the frequency and amplitude of the SOGI-FLL algorithm's input signal is modified in Ben Youssef and Sbita [13] for a system focused on the synchronization and connection of an inverter with the MN.

The application of LabVIEW and National Instruments cards has increased because of the benefits that they provide on the development of monitoring systems, and analysis and control of processes [14–20]. The LabVIEW software also presents benefits such as code reuse, high level of abstraction, and reduced time of design and validation [21]. The card PXIe-6356 and the software LabVIEW are used in Niknejad *et al*. [22] as a platform to control the speed of a DC brushless motor by manipulating a half bridge DC-DC converter. Because of the good parallel processing that a FPGA NI-sbRIO9606 provides, it is used together with LabVIEW in Dadu *et al*. [23] to validate experimentally a scheme for the control of a three-phase four-branches inverter. Because of the flexibility to add new sensors to the system, the Arduino card and the LabVIEW software are used in Montes *et al*. [24] for the automatic control of real time measurements for the plotting of the current-voltage curve of a photovoltaic array. Because of the high processing speed, the FPGA cRIO is frequently used for the control of electronic power converters [25, 26].

This paper presents the design of SOGI-FLL algorithm which is embedded on the NI myRIO platform using the LabVIEW virtual instrumentation software and is implemented on a DC-AC bidirectional converter to synchronize a DC Microgrid (MG) with the MN. The control algorithm is designed to maintain the stability of the DC bus by using the energy produced by the PA, extracting energy from the MN, or suppling energy to the MN, depending on the energy demand of the loads connected to the DC bus or of state of charge the battery bank. Non-linear loads such as computers, commercial fluorescent lamps, and LED, are used with the aim to evaluate the behavior of the system during real time situations. Additionally, the result section shows the systeḿs behavior, stabilization time and DC bus voltage, on load changes. Lastly, comparison with similar proposal found in literature are given in the last section of this paper.

## 2. DC MICROGRID STRUCTURE

All the schematic diagram of the MG is shown in Figure 1(a) and the complete experimental prototype is shown in

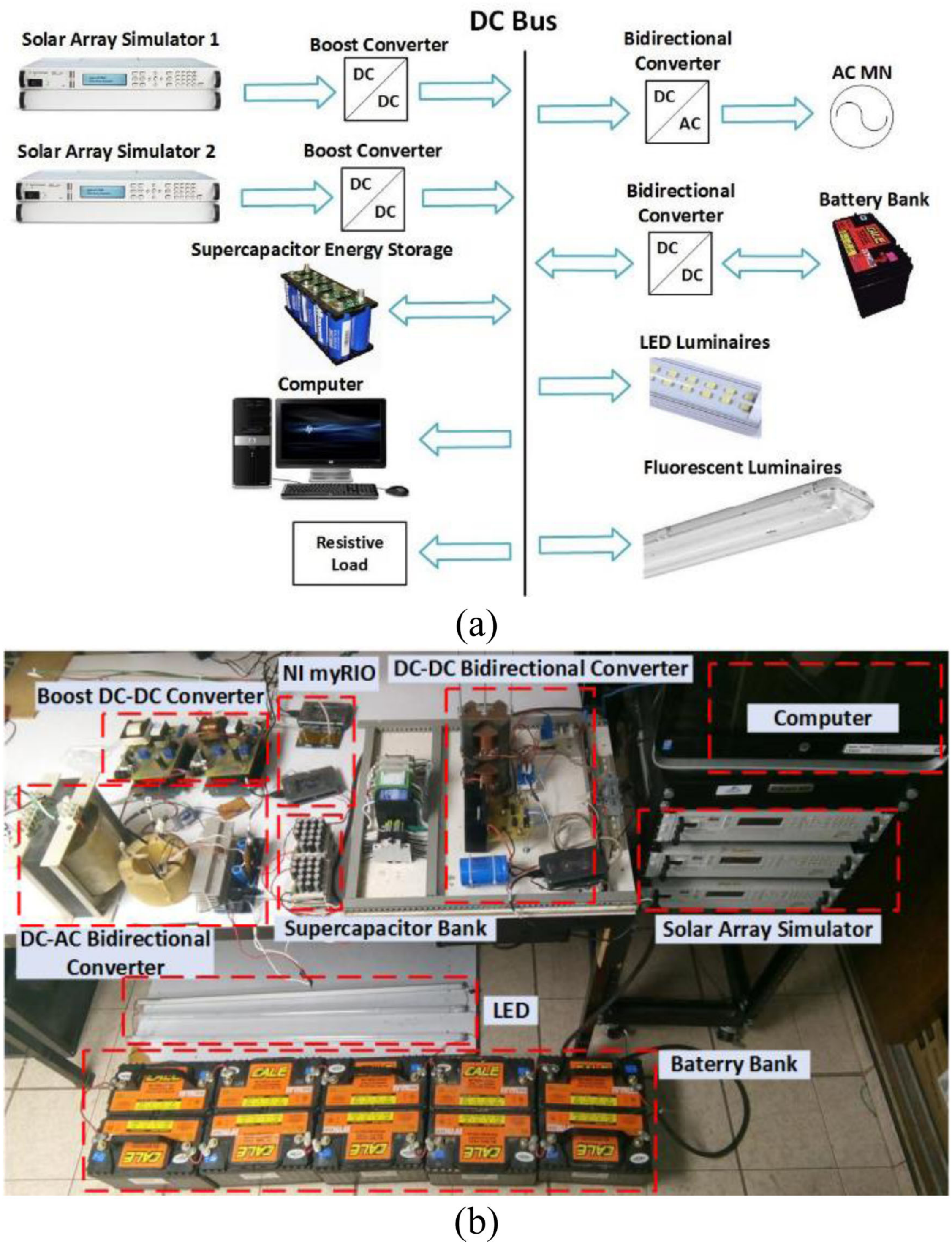



**FIGURE 1.** DC Microgrid: (a) schematic diagram and (b) view of the experimental prototype.

Figure 1(b). The MG consists of a 1 kW MG with a 190 V DC bus ($V_{\mu G}$). The energy is produced by two PA, each connected to a 0.6 kW boost DC-DC converter. The performance of the PA is emulated by two E4360A Agilent Solar Array Simulators (SAS). A bank of ten 115 Ah and 120 V lead-acid batteries is connected to the DC bus. Also, a 0.230 F bank of supercapacitors is connected in parallel to the DC bus with the purpose to maintain the stability of

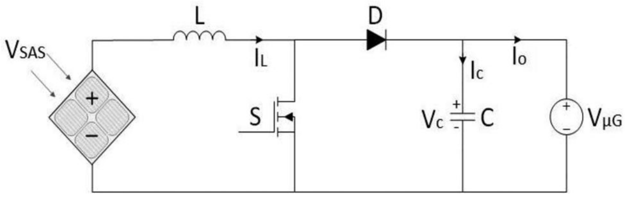



**FIGURE 2.** Circuit diagram of the boost DC-DC converter.

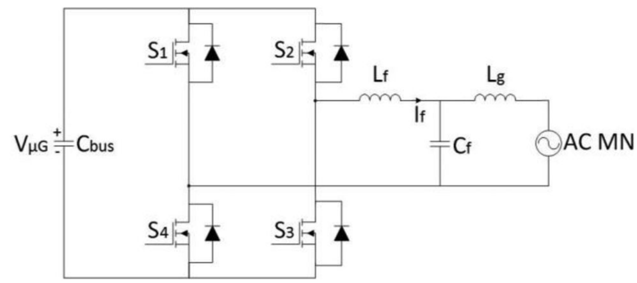



**FIGURE 3.** Circuit diagram of the DC-AC converter.

the DC bus voltage during the connection and disconnection of the non-linear loads (computer, resistive load, LED, and fluorescent lamps). This supercapacitor bank mitigates the transient voltage overshoots during load variations, but it also increases the settling time, as reported in the result section.

The MG is connected to the MN using a DC-AC bidirectional converter, which allows the transfer of energy between the MN and the MG. The energy that is not used by the loads and the bank of batteries is transferred from the MG to the MN, the DC-AC bidirectional converter works as an inverter, and when energy is transferred from the MN to the MG to complement the energy of the PA the DC-AC bidirectional converter works as a rectifier. In both cases, the control stabilization criteria were selected to ±5% of the DC-link voltage (190 V). It is important to note that other control proposals reported in literature achieved an error up to 10% [27].

### 2.1. Boost DC-DC Converter

Figure 2 shows the circuit diagram of the Boost DC-DC converter designed to obtain a voltage gain of $V_{\mu G}/V_{SAS} = 1.5$. This voltage gain guarantees that the output voltage of the converter is 190 V when the PA are at their maximum power point. The converter is designed to perform in a continuous conduction mode (CCM). In agreement with [28], for a frequency of 40 kHz, the values of $L$ and $C$ are 107 μH and 440 μF, respectively. The performance curve of the SAS is configured using a maximum power point voltage ($V_{\mathrm{mpp}}$) of 120 V, an open circuit voltage ($V_{\mathrm{oc}}$) of 130 V, a maximum power point current ($I_{\mathrm{mpp}}$) of 4.2 A, a short circuit current ($I_{\mathrm{sc}}$) of 5 A, and a maximum power of 500 W for each SAS.

### 2.2. DC-AC Bidirectional Converter

Figure 3 shows the circuit diagram of the DC-AC bidirectional converter used in the MG prototype. The energy transferred by the bidirectional converter is carried out by the power switches $S_1$, $S_2$, $S_3$ and $S_4$, these are in charge of controlling the amplitude and the frequency of the sine signal when the converter works as an inverter, in addition to control the energy that is transferred from the MN to the MG when it works as a rectifier. An *LCL* filter works as an interphase between the converter and the MN to set a sinusoidal wave according to the Official Mexican Norms (NOM) and the Mexican Electricity Commission (CFE). A capacitor, $C_{bus}$, is used on the DC bus to maintain the stability of the DC bus voltage. The minimum capacitance needed to assure that the DC bus voltage is maintained stable is obtained as

$$C_{bus} = \frac{(4P_{\mathrm{out}})(t_1)}{V^2_{\mu\mathrm{G,\ min}}} = 308\ \mu\mathrm{F} \tag{1}$$

where $P_{out}$ represents the output power, $V_{\mu\mathrm{G,min}}$ the minimum DC bus voltage required to transfer energy to the MN, and $t_1$ the time interval given as

$$t_1 = \frac{1}{4f_{\mathrm{grid}}} = 4.16\ \mathrm{ms} \tag{2}$$

where $f_{\mathrm{grid}}$ is the MN frequency, with a value of 60 Hz.

The filtering inductance is obtained as

$$L_f = \frac{(V_{\mu\mathrm{G}} - V_{\mathrm{grid,\ pk}})D}{n \cdot \triangle I_{\mathrm{f}} \cdot f_{\mathrm{sw}}} = \frac{(190-170)(0.8)}{(3)(0.1)(25000)} = 2.13\ \mathrm{mH} \tag{3}$$

where $n$ represents the number of levels of the inverter ($+V_{\mu\mathrm{G}}$, $-V_{\mu\mathrm{G}}$, and $0$), $D$ the duty cycle, $f_{\mathrm{sw}}$ the commutation frequency, and $\Delta I_f$ the ripple current. The criteria considered for the design of the filtering inductance is that $\Delta I_f$ should be 10% of the nominal value of $I_f$.

The criteria for the design of the capacitance, $C_f$, is that it should be able to allow the exchange of the maximum reactive power. The design conditions for the capacitor are

$$X_{C_f} \geq \frac{V^2_{\mathrm{grid,max}}}{P_{\mathrm{reactive}}} = 54\ \Omega \tag{4}$$

$$C_f \leq \frac{1}{\omega \cdot X_{C_{\mathrm{f}}}} = 49\ \mu F \tag{5}$$

The filter resonance frequency, $f_{res}$, is given as

$$f_{\mathrm{res}} = \frac{1}{2\pi}\sqrt{\frac{L_{\mathrm{f}} + L_{\mathrm{g}}}{(L_{\mathrm{f}})(L_{\mathrm{g}})(C_{\mathrm{f}})}} \quad (6)$$

where $f_{\mathrm{res}}$ should be 0.5 times the commutation frequency and 10 times the fundamental frequency of the MN to avoid any problems related with the resonance state of the *LCL* filter.

This criterion is met by assigning values of 1 mH and 1 μF to $L_{\mathrm{g}}$ and $C_{\mathrm{f}}$, respectively, and considering a 2% of the value obtained for $C_{\mathrm{f}}$ [29]. As a result, the value for $f_{\mathrm{res}}$ is 6.1 kHz.

### 2.3. Linear and Non-Linear Loads

To evaluate the MG during typical load conditions, in this work was used linear and non-linear loads. The linear load was emulated with a resistor bank of 600 Ω, and the non-linear load was implemented with two different configurations. The first one was an array of three illuminating systems with two different technologies. This arrangement was constituted by seven 23 W LED lamps, four 28 W Louver lamps plus one electronic ballast. The second configuration was a desktop computer, which demanded a maximum power of 300 W. The main reason to use these typical applications was to prove that the proposed MG topology and control strategy, does not require an additional arrangement or process. Therefore, those proposal can be implemented in a real-life scenario.

## 3. CONTROL SYSTEM

The Control System (CS) is implemented on the experimental prototype of the MG to maintain the energy balance on the different elements of the MG, this has been divided in two Control Systems, the Control System 1 (CS1) and the Control System 2 (CS2), implemented in NI myRIO_1 and NI myRIO_2 platforms using LabVIEW, respectively (see Figure 4). CS1 and CS2 defines the mode of operation of the local control applied to each of the converters. Its main goal is to maintain the DC bus voltage in a range of 190 V ± 5% to assure that the loads connected to the DC bus work well. The DC-AC converter performs the following tasks:

1. When the maximum power of the PA is extracted and the voltage of the DC bus is lower than 182 V, it is necessary to transfer energy from the MN to the DC bus to keep it on the established range of operation. In this mode of operation, the DC-AC converter works as a rectifier, while the Boost DC-DC converter works with the algorithm for the MPPT. In this case, the energy from the MN can be transferred to the batteries bank.
2. When the DC-DC converters, supplied by the SAS, work on a voltage control mode, the energy produced by the SAS is no longer used. In this case, the CS commutes to the control algorithm of the MPPT on the SAS and activates the DC-AC bidirectional converter as an inverter, transferring the energy that is not used by the loads to the MN. In this case, the energy from the MG can be transferred to the batteries bank.

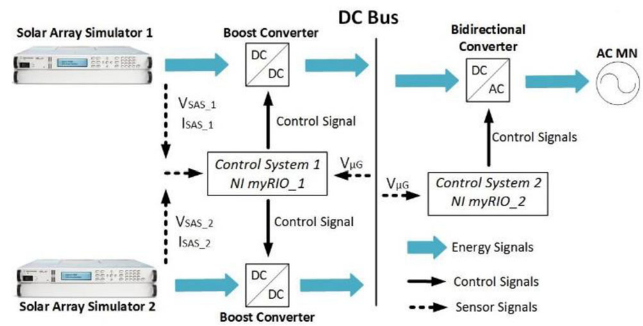



**FIGURE 4.** Block diagram of the implemented control system.

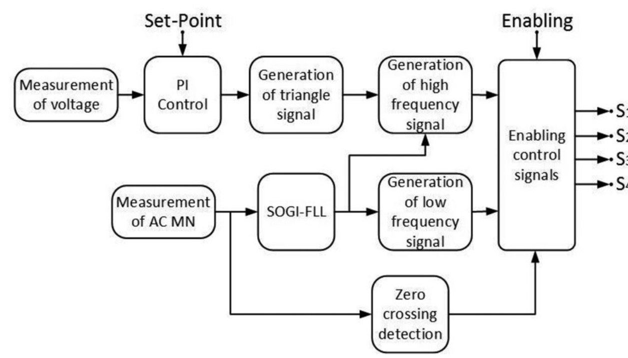



**FIGURE 5.** Block diagram of the control algorithm for the DC-AC bidirectional converter working as an inverter.

Figure 4 shows the block diagram of the control system applied to the experimental prototype of the MG. The CS1 controls the two DC-DC converters and works on two operation modes. The first mode uses a classic control action (PI control) to maintain the DC bus voltage level between the defined limits. In this operation mode, the battery bank and main grid are off. The second mode uses an algorithm based on the P&O (Perturb and Observe) algorithm to control the MPPT. With the CS1, the voltage and current delivered by the SAS are permanently monitored during the operation of the prototype.

Figure 5 shows the block diagram of the CS2 general structure working as inverter. This system has as output signals $S_1$, $S_2$, $S_3$ and $S_4$ which control the DC-AC bidirectional converter to transfers energy from the DC bus (MG) to the MN, and vice-versa (those signals were activated with the enabling signal from CS1). These signals are generated by comparing the output signal of the SOGI-FLL with a 25 kHz triangular signal. The amplitude of the triangular signal is modified based on the DC (190 V) reference bus voltage and the measured one. If the DC bus voltage increases, the amplitude of the triangular signal increases, allowing the transferring of the maximum amount of energy to the MN. If the DC bus voltage decreases, the amplitude of the triangular signal decreases, allowing the transferring of the minimum amount of energy to the MN. These actions stabilize the DC bus voltage on the defined value.

Figure 6 shows the circuit diagram of the DC-AC bidirectional converter. When energy is transferred from the DC bus to the MN the DC-AC bidirectional converter works as an inverter, and the CS2 maintains the DC bus stable using a PI control action. If the DC bus voltage level decreases, the energy that is transferred to the MN also decreases, which is reflected in an increase of the DC bus voltage level. If the DC bus voltage level increases, the energy that is transferred to the MN also increases, which is reflected in a decrease of the DC bus voltage level. The synchronization between the signal generated by the DC-AC bidirectional converter working as an inverter and the MN is obtained by the zero crossing detection modules and the SOGI-FLL algorithm. The SOGI-FLL technique uses the SOGI structure together with a frequency locked loop (FLL). The SOGI-FLL structure is shown in Figure 7, *V* is the measured reference signal of the MN, the output signal *V'* has the same phase and magnitude as the signal *V*, while the signal *Qv'* is offset 90 degrees with respect to the measured signal, *W'* is the frequency of resonance, *eff* is the control signal of the SOGI algorithm. This technique is based on the estimated frequency *W'* instead of the estimated phase *θ'*, as opposed to those synchronization techniques based on PLL. Thus, the SOGI-FLL technique does not need a voltage-controlled oscillator to adjust the frequency of the reference signal. This is because the SOGI-FLL technique needs one central frequency of oscillation only, which makes the digital implementation of the SOGI-FLL very practical, and it reduces substantially the number of components compared to other synchronization techniques such as EPLL, QPLL, SOGI-PLL and SOGI-FLL [30]. Also, the SOGI-FLL technique provides a fast and accurate frequency tracking, under harmonic distortion, phase change, and frequency and amplitude variation.

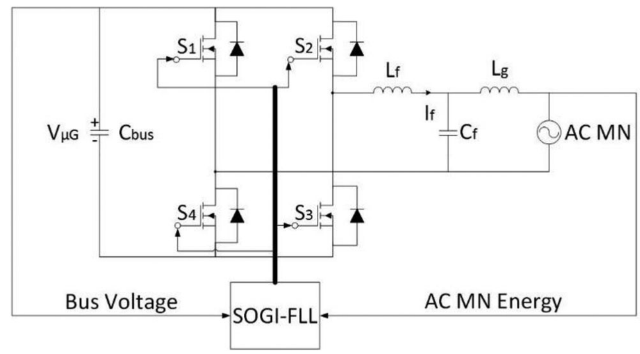



**FIGURE 6.** Circuit diagram of the DC-AC bidirectional converter working as an inverter.

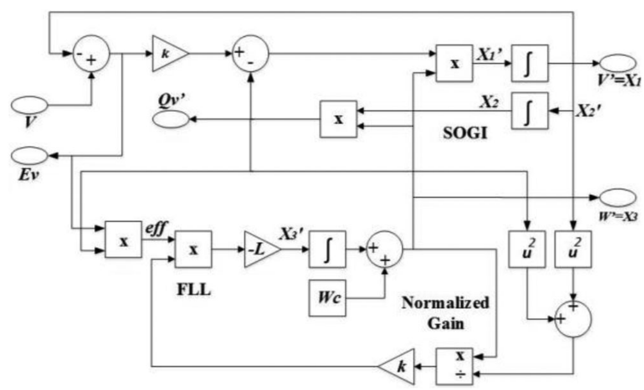



**FIGURE 7.** Schematic representation of the SOGI-FLL structure.

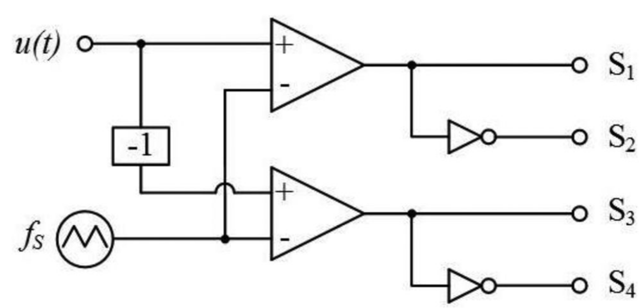



**FIGURE 8.** Diagram of the control system for the DC-AC bidirectional converter working as a rectifier.

If the DC-AC bidirectional converter works as rectifier, the control system is done with a classic PI controller. This controller stabilizes the DC bus at a voltage level of 190 V, by increasing and decreasing the amplitude of the high-frequency triangular signal, $f_s$, which is compared with a low-frequency sinusoidal signal, *u(t)*, as is show in Figure 8.

Figure 9 shows the control algorithm used for the monitoring of the MN, the zero-crossing detection of the MN,

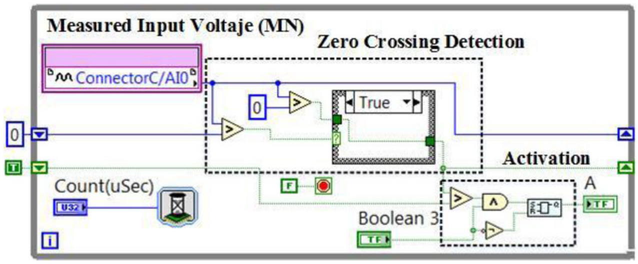



**FIGURE 9.** Monitoring of the MN, detection of zero crossing, and activation of the control algorithm.

and the activation of the control for the DC-AC bidirectional converter working as an inverter. The "Boolean 3" control signal (enabling signal Figure 5) is generated by the CS, when this signal is activated, the algorithm synchronizes the zero crossing of the MN and activates the "A" variable which activates the pulses that are sent to the power interrupters $S_1$, $S_2$, $S_3$ y $S_4$ of the converter shown in Figure 6. The average time between obtaining a sample of the inlet and emitting a control action is 10 µs.

Figures 10(a) and 10(b) show the algorithm used to generate the triangular signal with 25 kHz frequency. The first section generates a ramp with a positive slope of 0 to 20 and a time interval of 40 tick, Figure 10(a). Each tick corresponds to 25 ns because a 40 MHz FPGA oscillator was used. The second section generates a ramp with a negative slope of 19 to 0, Figure 10(b). The "y2" control signal manages the amplitude of the triangular signal that comes from the control output of the classical PI algorithm.

The high-frequency "triangular" output signal is compared with the sinusoidal output signal of the SOGI-FLL algorithm called "AC Ref" to generate the "Boolean 2" signal (see Figure 11). The "Boolean 2" signal is used for the high frequency control of the two power interrupters of the converter working as an inverter. In this structure, the high-frequency control signals are enabled by the signal control "A" from Figure 9, and the digital lines *DIO3* ($S_1$ control signal) and *DIO0* ($S_3$ control signal) of the *C port* of the NI myRIO-1900 card are used.

The output signal "AC Ref" of the SOGI-FLL algorithm is used to generate the low-frequency signals for the control of the full bridge inverter, obtaining a digital signal called "Boolean" (see Figure 12) which is in phase with the "AC Ref" signal. This "Boolean" signal is sent to an algorithm to generate a complimentary signal for "Boolean" and to add a dead time of 225 ns between the two signals. Figures 13(a) and 13(b) show the algorithm used to generate the dead time between the low frequency control signals of the full bridge inverter. The outputs of the low frequency control signals of the full bridge inverter are the *DIO1* ($S_2$ control signal) and *DIO2* ($S_4$ control signal) of *port C* of the NI myRIO card. The hardware

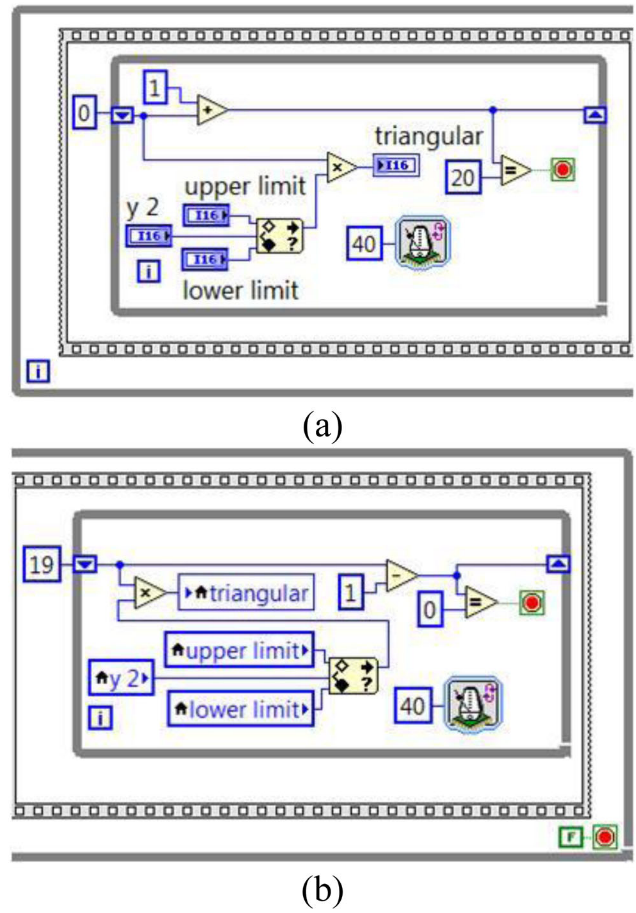



**FIGURE 10.** Generation of a 25 kHz triangular signal: (a) Positive slope and (b) Negative slope.

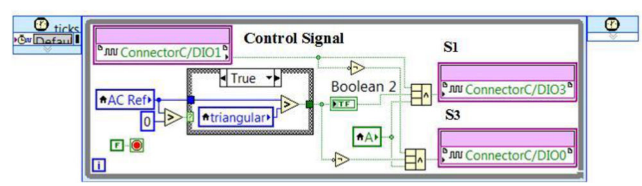



**FIGURE 11.** Comparison of the sinusoidal and triangular signals, and activation of the high frequency signals.

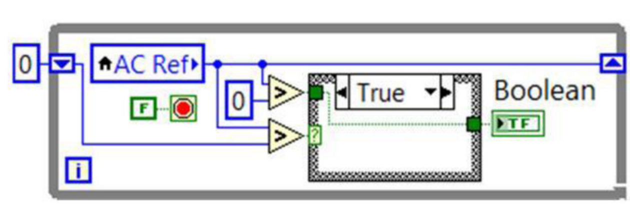



**FIGURE 12.** Generation of low frequency control signals.

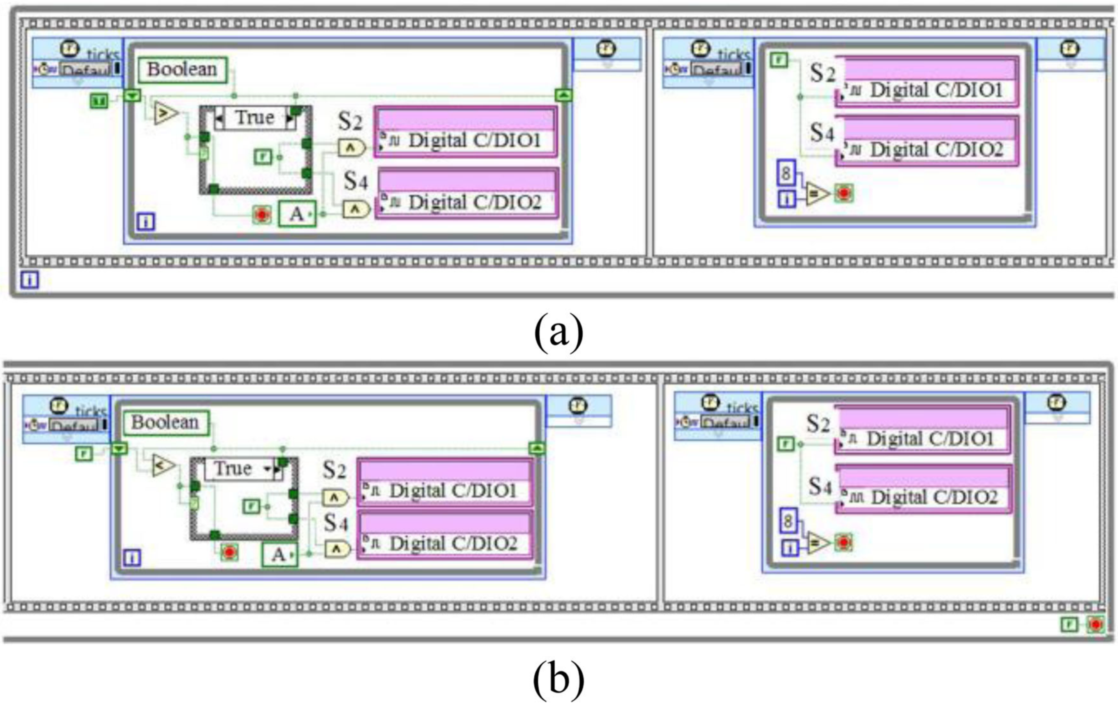



FIGURE 13. Algorithm for the generation of dead time.

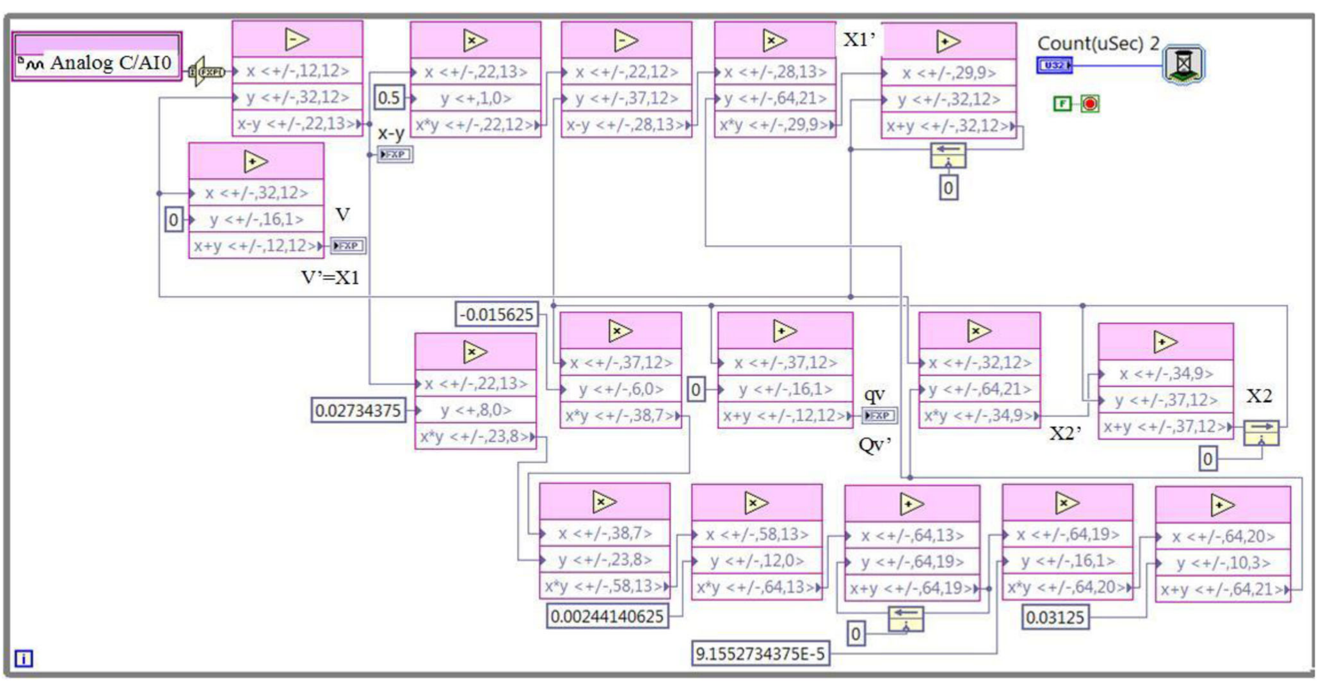



FIGURE 14. SOGI-FLL algorithm.

implementation of the normalized SOGI-FLL algorithm is shown in Figures 14 and 15. Figure 14 shows the design of the SOGI-FLL algorithm, while Figure 15 shows the normalization. This stage allows to obtain a sinusoidal signal with constant amplitude and frequency parameters, regardless of the perturbations in the measured signal, Figure 23(a).

Labels are used in both figures to relate the design of the algorithm in LabVIEW with the block diagram of the synchronization algorithm that is shown in Figure 7. The

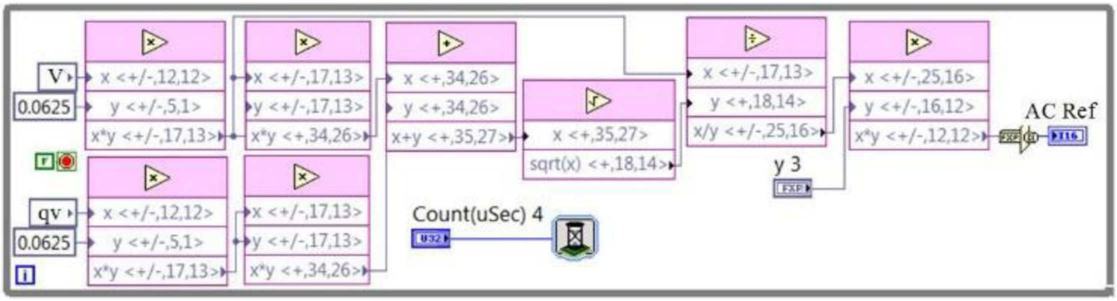



**FIGURE 15.** Normalization of the SOGI-FLL algorithm.

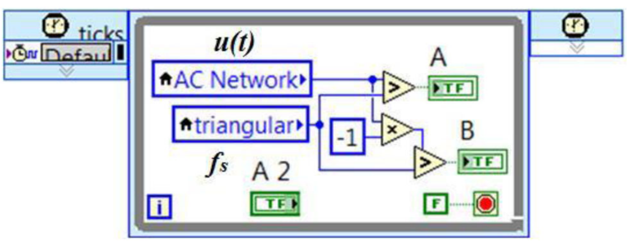



**FIGURE 16.** Control algorithm for the rectifier.

execution time of each iteration of the SOGI-FLL algorithm is 86 μs.

Figure 16 shows the implementation in LabVIEW of the control algorithm for the DC-AC converter working as a rectifier (see Figure 8). This algorithm compares the *u(t)* signal with the 12.5 kHz triangular signal $f_s$ to obtain the control signals "A" and "B" used to control the power interrupters of the DC-AC converter, $f_s$ is internally generated in the myRIO by using an algorithm similar to Figure 10.

After the implementation of the algorithm shown in Figure 8, it is necessary to obtain the complimentary signals and assign the dead time to the outputs of each of the signals that are assigned to the power interrupters. To this end, it is used a similar logic shown in Figures 13(a) and 13(b), with a dead time of 12 ns.

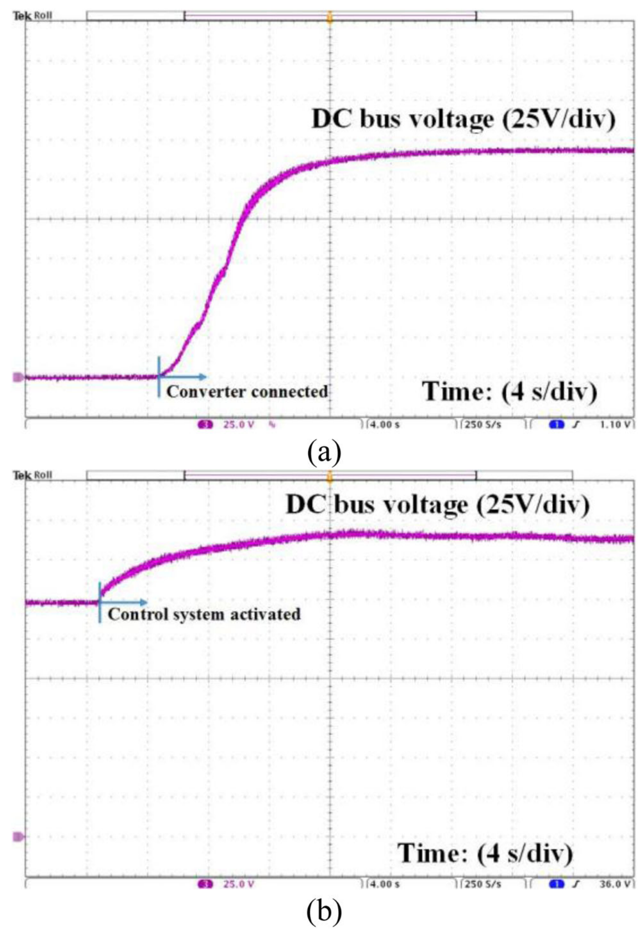



**FIGURE 17.** DC bus behavior: (a) connection of the DC-AC bidirectional converter with the MN, and (b) activation of the control system.

## 4. RESULTS

For the first experiments developed with the MG prototype and the embedded control system, the DC-AC bidirectional converter is used as rectifier supplying energy to the DC-link. Figures 17–24 were acquired with a Tektronix MDO4054C, the horizontal axis means time, and vertical axis means voltage or current, as indicated in the bottom of each figure.

Figure 17(a) shows the behavior for the DC bus (purple line) when the DC-AC bidirectional converter, with a voltage of 144 V, is connected with the MN. As it can be noticed, the DC voltage reaches the steady state of 144 V at 16 sec. This time is defined by the supercapacitor value, in this case 0.23 F. The supercapacitor main function is to reduce the DC link voltage droop and transient current demand during load changes or sources connection/

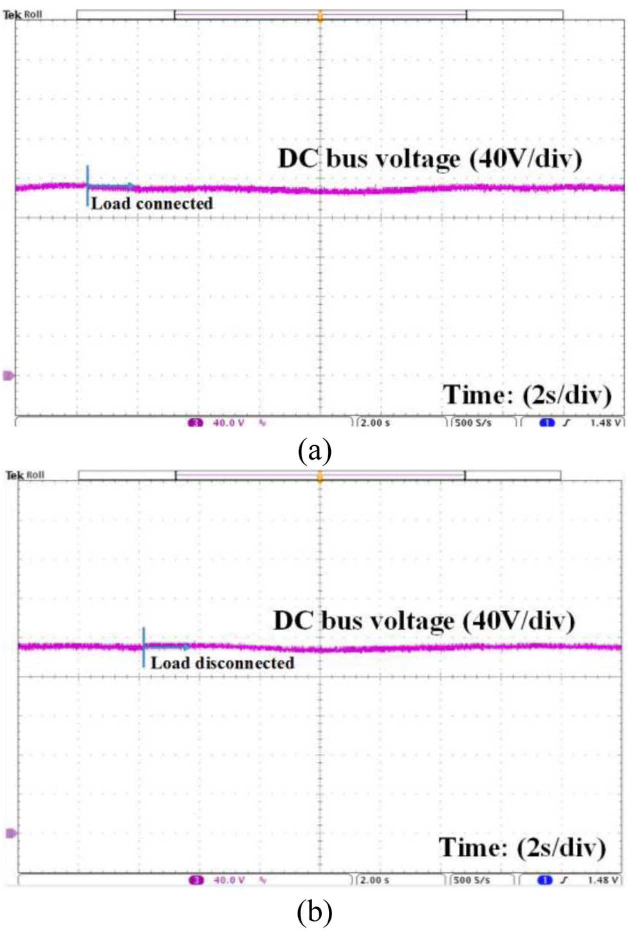



**FIGURE 18.** DC bus behavior with a 125 W LED lamp: (a) connection and (b) disconnection.

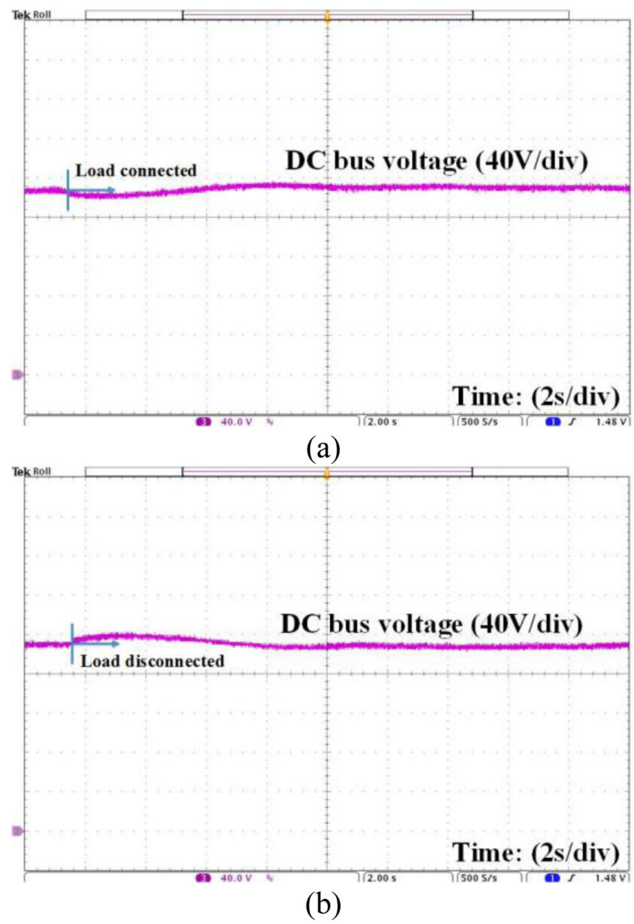


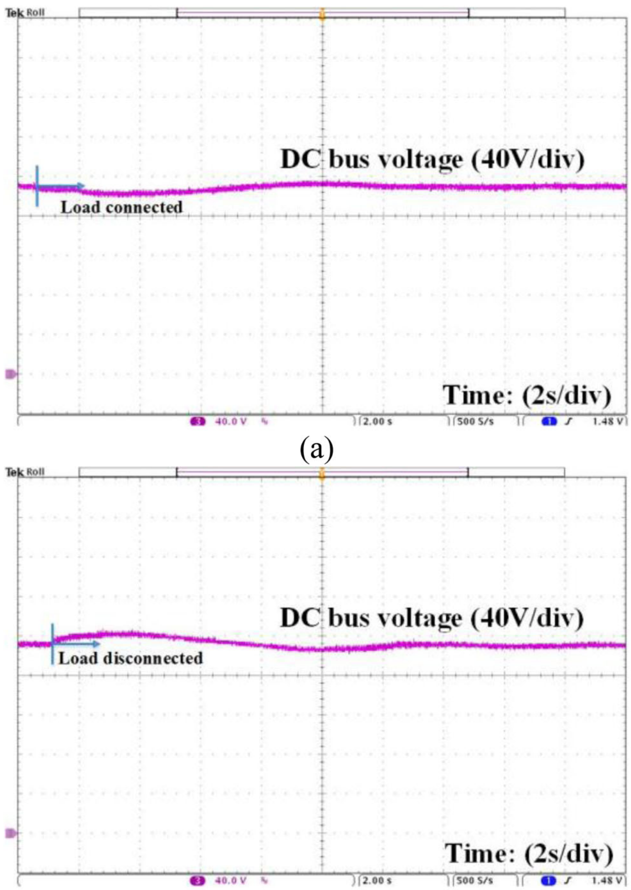

**FIGURE 19.** DC bus behavior with a 300 W fluorescent lamp: (a) connection and (b) disconnection.

disconnection. Figure 17(b) shows the behavior for the DC bus (purple line) when the control system in charge of maintaining the stability of the DC bus voltage is activated. The stability (settling time) is reached 14 s after the controller is on. Once the DC bus is stabilized, the non-linear loads are connected and disconnected to the DC bus.

Figure 18(a) shows the behavior for the DC bus (purple line) when a 125 W LED lamp is connected to the DC bus, and Figure 18(b) shows the behavior for the DC bus (purple line) when a 125 W LED lamp is disconnected.

It was observed that the DC bus suffers a voltage droop and an overshoot of 1.5%. This behavior is due to the load connection/disconnection, which is reduced by the supercapacitor bank. Figure 19a shows the effect on the DC bus (purple line) when a 300 W fluorescent lamp is connected to the DC bus, and Figure 19b shows the effect on the DC bus (purple line) when a 300 W fluorescent lamp is disconnected from DC bus. We observed that when the 300 W fluorescent lamp is connected to the DC bus, the nominal voltage is decreased by 3.15%, and when the 300 W fluorescent lamp is disconnected from the DC bus, there is an overshoot of 3.15%. The DC bus is stabilized at 6 s.

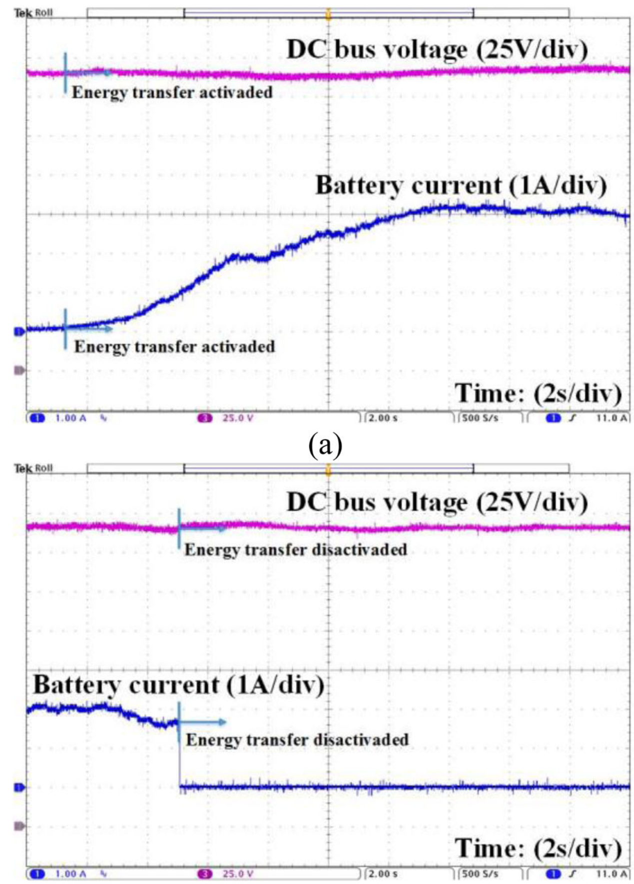
Figure 20(a) shows the behavior for the DC bus (purple line) when a 400 W fluorescent lamp is connected to the DC bus, and Figure 20(b) shows the behavior for the DC bus (purple line) when a 400 W fluorescent lamp is disconnected from the DC bus. It was observed a decrease of 3.6% in the nominal voltage when the load is connected, and an overshoot of 3.6% when the load is disconnected. The time required to stabilize the DC bus was 7 s. The experimental prototype also allows the transfer of energy from the MN to the DC MG using the DC-AC bidirectional converter as a rectifier in order to transfer the energy to the batteries bank. Figure 21(a) shows the behavior of the DC bus voltage (purple line) and the current (blue line) that is injected to the bank of batteries. In this operation mode, the bidirectional DC/DC converter behaves like Buck



(a)

(b)

**FIGURE 20.** Behavior for the DC bus with a 400 W fluorescent lamp: (a) connection and (b) disconnection.



(a)

(b)

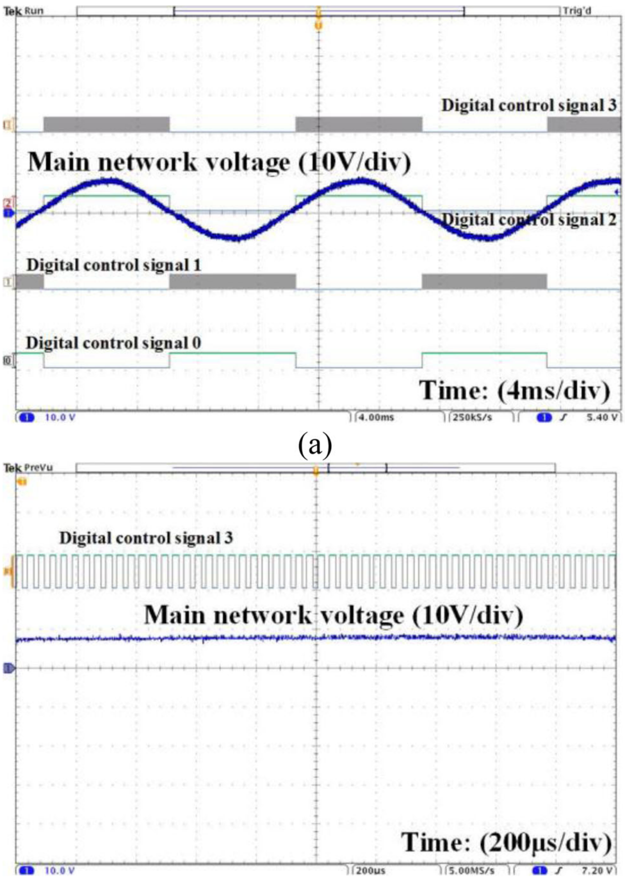
**FIGURE 21.** DC bus voltage (purple line) and battery bank current (blue line) for the: (a) transferring and, (b) stop transferring of energy from the MN to the batteries bank.

converter and it charges the battery bank. Once the current injected to the batteries bank is increased, the DC bus suffers a voltage reduction of 1%. During this transition, a soft commutation is defined (12 s); this last to avoid an undesirable converter stress and significant DC voltage drop.

Once the battery bank is charged up to 80%, the current supply is stopped and the system changes to constant voltage mode to complete the charging of the batteries. Figure 21(b) shows the behavior of the DC bus (purple line) when the transfer of energy from the MN to the bank of batteries is stopped. This behavior is like a load bus disconnection, which generates an overshoot of 1.3%. In both cases, the rectifier works with the embedded system described previously.

For the operation of the battery bank, the operation states are defined based on the following criteria: (1) the charge and discharge processes of the battery bank must always be concluded, (2) the continuous current charge process is completed once the battery bank achieves a maximum voltage value of 135 V, (3) the continuous charge voltage process is completed when the current injected into the battery bank is less than 0.5 A, and (4) the discharge process is completed when the battery voltage is 100 V. It should be noted that the bidirectional power converter used in the battery bank has a maximum capacity of 1 kW, for which reason the maximum current that can be extracted or injected into the bank is 10 A.

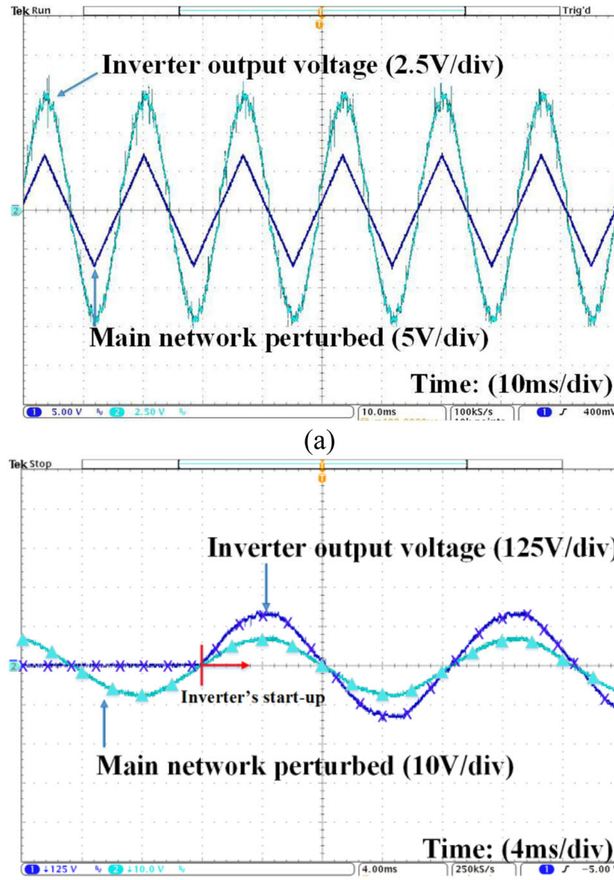
Control signals, and input and output signals of the DC-AC bidirectional converter working as an inverter are also analyzed on the MG. Figure 22(a) shows the monitoring of the MN (blue line) and the generation of the low and high frequency pulses (green line) used for the power interrupters of the inverter. It is observed that the low frequency pulses are synchronized with the MN signal. Also, it is observed that the control signals "0" and "1" are activated in the negative part of the sinusoidal signal of the MN, and the control signals "2" and "3" remain off. The opposite is observed for the positive part of the sinusoidal signal of the MN, the control signals "0" and "1" remain off, and the control signals




(a)

(b)

**FIGURE 22:** (a) Monitoring of the MN (blue line) and control signals (green line) of the inverter, and, (b) behavior of the high frequency control signal.




(a)

(b)

**FIGURE 23.** Generating the sinusoidal signal: (a) response of the control system to a perturbation on the MN, and (b) inverter's startup at the zero crossing of the MN.

"2" and "3" are activated, generating the output signal of the inverter that is synchronized with the MN.

Figure 22(b) shows the behavior of the high frequency control signal (green line), showing specifically the variation of the wide of the pulse in charge of the generation of the sinusoidal signal (blue line) that is synchronized with the MN. One of the main characteristics of the SOGI-FLL algorithm is that, if the MN suffers perturbations (blue line), the control system is in charge of generating the sinusoidal signal (green line) at the exit of the inverter, as shown in Figure 23(a). Figure 23(b) shows the performance of the inverter's startup (blue line), which begins with the zero crossing of the MN, changing from the negative part to the positive part of the sinusoidal signal (green line).

Variations on the amplitude of the high frequency triangular signal cause a modification to the amplitude of the sinusoidal signal at the output of the inverter (blue line). Figures 24(a) and 24(b) show the effect of increasing the amplitude of the triangular signal, which causes an increase of 30 V on the output of the inverter. If the inverter is connected to the MN, this action generates a bigger energy transfer to the MN. Figure 25 shows the current and voltage parameters at the output of the inverter when it is connected to the MN. The non-linear loads (LED and fluorescents) are connected to the DC bus to modify the amount of energy that is transferred to the MN. The main objective of these tests was to show the current and voltage steady state behavior for different power values. It was observed that the proposed system compensates the active and reactive power once a load variation occurs.

Table 1 shows the transient response during connection and disconnection state between loads and power sources. It is necessary to mention that DC-link voltage's dynamic is slow, as mention before, due to safety concerns. Indeed, the supercapacitor bank spends 16 sec from 0 V to 144 V, Figure 17a. Table 2 shows the use of resources from the

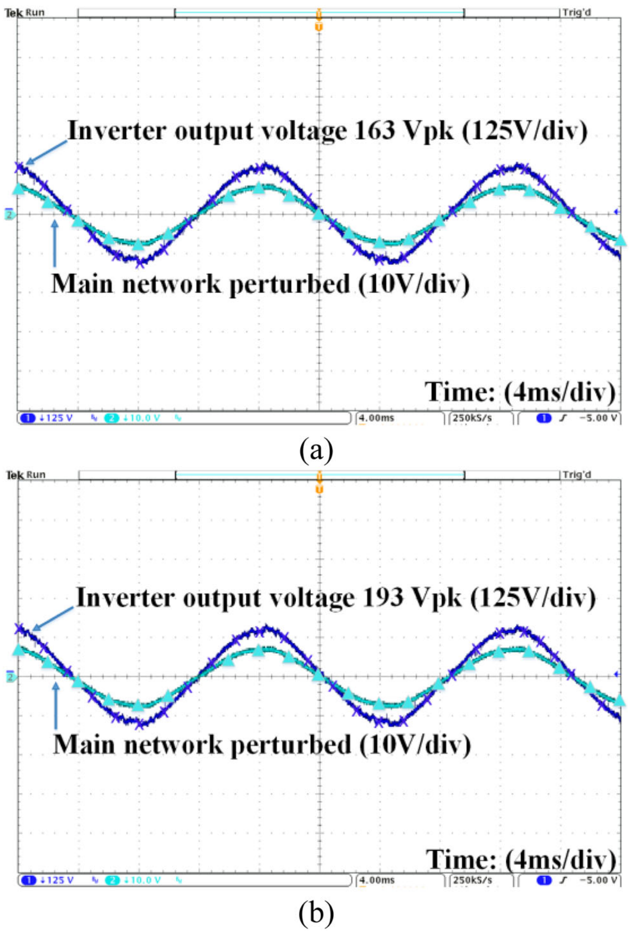



**FIGURE 24.** (a, b) Effects generated by modifying the amplitude of the triangular high frequency signal.

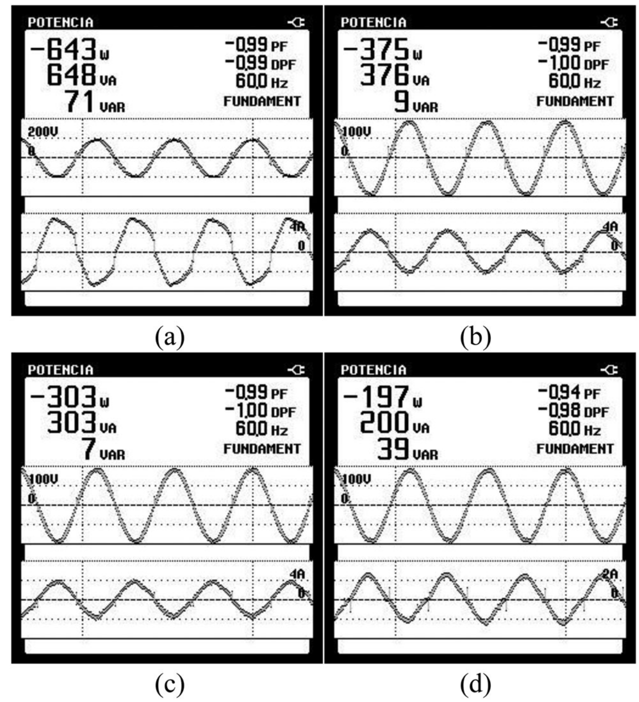



**FIGURE 25.** (a–d) Behavior of the MG's voltage and current when non-linear loads are connected to the MN.

| Test | Settling time | Perturbation |
|---|---|---|
| Control activated | 14 sec | 0% |
| 150 W load | 4 sec | 1.5% |
| 300 W load | 6 sec | 3.15% |
| 400 W load | 7 sec | 3.6% |
| Battery connected | 12 sec | 1% |

**TABLE 1.** DC link behavior.

| Device utilization | Used | Total | Percent (%) |
|---|---|---|---|
| Total slices | 3422 | 4400 | 77.8 |
| Slice register | 9770 | 35200 | 27.8 |
| Slice LUTs | 10041 | 17600 | 57.1 |
| Block RAMs | 5 | 60 | 8.3 |
| DSP48s | 34 | 80 | 42.5 |

**TABLE 2.** Use of resources from the FPGA.

FPGA by the implementation of the embedded control algorithms used on the DC-AC bidirectional converter. This control algorithm uses 77.8% of the total slices due to the arithmetic operations are embedded by the DSP48s blocks, which avoids exhausting the resources of the FPGA.

## 5. COMPARISON WITH OTHER TECHNIQUES

It is difficult to find a similar experimental prototype as the one presented in this work to make a one to one comparison. However, in Sun *et al*. [27] a distributed control strategy is presented for the behavior of a DC bus of a MG with a 900 W modular photovoltaic generation.

The modes of operation of the control system are defined based on the voltage bus level, which works in a voltage range of 180 V to 210 V. However, during transient periods between operation modes, the operation voltage is 190 V, ±10%.

Two 500 W DC-AC bidirectional converters and only resistive loads are used in Sun *et al*. [27]. Results show that the bus works with a voltage of 190 V ± 10%. Overshoots of 14% the nominal voltage are observed on the DC bus with stabilization times of 1 sec. These overshoots are bigger than the ones obtained with the

| Reference | Response time | Perturbation |
|---|---|---|
| Sun *et al.* [27] | 1 sec | 14% |
| Zhang *et al.* [31] | 1 sec | 6% |
| Dagbagi *et al.* [12] | 150 msec | 7.5% |
| Our proposal | 6 s | 3.6% |

**Table 3.** DC link voltage comparison.

experimental prototype presented in this work. Indeed, the battery bank's current presents a noise significant overshoot of 60%, with a settling time of 1 sec.

Another control algorithm for a DC MG is presented in Zhang *et al.* [31]. The DC MG works with a DC bus in a range of voltage of 180 to 210 V. The prototype shows overshoots and voltage drops of 6%, and noise on the frequency of the DC bus, which is generated by the interconnection with the MN. Another similar study to the one developed in the present work is shown in Dagbagi *et al.* [12]. A real-time embedded control system is applied to a three-phase rectifier connected to the MN. The DC bus works with a voltage of 200 V. The experimental results show that the control system maintains the stability of the DC bus when a 100 Ω, 2.5 a resistive load is connected or disconnected. Also, voltage drops of 7.5% and overshoots of 6.25% are observed on the DC bus, with response times near to 150 ms.

A similar control technique to the one presented in this paper was also reported in Shamshuddin *et al.* [32]. Although the MG was isolated from the MN, the proposed controller extracts the maximum power available from the PV and fuel cell to achieve a feasible optimum battery bank charge. Table 3 summaries the overall response time and those compared with this proposal (resistive load). As it can be noticed from Tables 1 and 3, although, this paper proposal reports stabilization times of 6 s, which is significant compared with previous works. It also reported the lowest perturbation variance during load transient (overshoots of 3.6%), which makes it attractive for sensitive application; and it allows to visualize the advantage of this proposal.

## 6. CONCLUSION

This paper presents the design and implementation of an embedded control algorithm for a DC-AC bidirectional converter used for the interconnection of a MG with the MN. The main purpose of the embedded control system is to maintain the stability of the DC bus of the MG at a given value of 190 V ± 5% when the DC-AC bidirectional converter controls the energy management of the MG.

The use of the NI myRIO-1900 platform allows designing the embedded control system so that the bidirectional converter works in both operation modes. This platform shows three advantages which are (1) the integration of the analog converters used to monitor the MN and the DC bus of the MG, (2) the including the high-speed output signals used to control the four power switches that modify the operating mode of the converter and the amount of energy transferred, and (3) the versatility of programing using LabVIEW virtual instrumentation software. When generating the embedded controller, the use of the DSP48s blocks allows integrating both control algorithms in the same device without exhausting the resources of the FPGA (Slices 77.8%, Registers 27.8%, LUTs 57.1%, block RAMs 8.3% and DSP48s 42.5%).

The reported tests scenarios were three (1) the connection and disconnection of different non-linear loads and sources to the DC bus, (2) the transfer of energy from the MN to the MG, and (3) the supply of energy from the MG to the MN. The energy sources were two photovoltaic arrays, a batteries bank, and the loads correspond to computers, commercial fluorescent lamps, and LED. The experimental results show that the embedded control system is capable of maintain the stability of the DC bus when it interacts with the different non-linear loads connected to the DC MG.

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